

WAUTOMA, WI, USA

WMO: 720398

Lat: 44.042N

Lon: 89.304W

Elev: 262

StdP: 98.22

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.0	(c) -22.0	(d) -27.9	(e) 0.3	(f) -22.9	(g) -25.9	(h) 0.4	(i) -21.4	(j) 10.9	(k) -10.0	(l) 9.4	(m) -10.1	(n) 1.9	(o) 290	(p) 0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	32.5	22.5	30.9	22.1	28.9	21.1	24.5	29.7	23.5	28.4	22.5	27.0	4.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.1	26.7	22.2	17.4	26.0	21.2	16.4	24.9	75.3	30.2	71.1	28.3	67.5	27.3	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a) 9.3	(b) 8.2	(c) 7.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	
(a) 9.3	(b) 8.2	(c) 7.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-9.0	-6.9	0.0	6.9	13.9	19.2	21.7	20.3	16.0	8.9	1.6	-5.4	(6)
	DBStd	11.81	6.83	6.40	6.62	5.01	4.96	3.48	3.12	3.08	4.31	4.86	5.54	6.02	(7)
	HDD10.0	2339	589	474	317	120	17	0	0	0	4	81	258	478	(8)
	HDD18.3	4305	848	707	569	344	156	31	7	16	95	294	503	737	(9)
	CDD10.0	1363	0	0	7	27	137	276	361	319	184	48	4	0	(10)
	CDD18.3	288	0	0	0	1	17	56	110	76	25	3	0	0	(11)
	CDH23.3	3207	0	0	6	24	270	659	1181	724	294	47	0	0	(12)
	CDH26.7	1035	0	0	0	3	89	215	421	214	84	9	0	0	(13)
(14) Wind	WSAvg	2.8	3.2	3.0	2.8	3.6	3.3	2.6	2.3	2.1	2.3	2.8	3.0	3.0	(14)
(15) Precipitation	PrecAvg	819	32	29	45	80	101	119	98	100	80	60	46	35	(15)
	PrecMax	976	73	52	97	141	192	223	228	245	206	137	119	76	(16)
	PrecMin	665	13	12	7	27	49	44	32	33	15	11	1	7	(17)
	PrecStd	84	16	11	25	29	37	58	57	45	43	32	32	20	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	13.6	24.8	27.0	32.7	33.9	37.5	32.8	32.0	27.7	20.7	12.6	(19)
		MCWB	4.1	8.4	16.4	15.9	21.8	23.0	24.0	22.5	23.2	19.7	13.7	11.6	(20)
	2%	DB	4.2	7.2	18.0	22.7	28.9	31.5	32.5	31.1	29.0	23.8	16.6	7.3	(21)
		MCWB	2.1	4.4	12.3	14.0	19.1	22.1	22.8	22.4	21.7	17.6	12.1	5.4	(22)
	5%	DB	2.5	4.0	14.0	19.6	26.4	28.9	30.5	28.9	27.0	21.0	13.1	5.0	(23)
		MCWB	1.3	1.9	8.8	11.6	17.4	20.5	22.1	21.6	19.6	15.8	9.1	3.8	(24)
	10%	DB	1.2	2.1	10.0	17.1	23.5	27.2	28.8	27.4	24.1	17.8	10.1	2.7	(25)
		MCWB	0.4	0.4	6.2	10.4	16.0	19.3	21.3	20.7	18.5	13.9	7.3	1.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	9.1	17.7	18.0	22.8	25.2	25.8	25.2	24.2	21.9	15.1	10.9	(27)
		MCDB	7.0	13.9	24.3	23.0	30.9	30.6	32.0	30.0	30.0	26.3	18.9	11.5	(28)
	2%	WB	2.8	4.9	14.3	15.3	20.8	23.5	24.6	24.0	22.7	18.8	12.6	5.8	(29)
		MCDB	4.2	7.3	17.8	20.9	26.6	28.9	30.1	28.4	27.2	21.4	16.3	6.4	(30)
	5%	WB	1.5	2.3	9.7	13.1	19.4	22.2	23.6	22.8	21.3	16.7	9.9	3.6	(31)
		MCDB	2.2	4.0	13.8	18.1	23.9	26.8	28.6	26.7	24.5	19.7	13.1	4.8	(32)
	10%	WB	0.2	0.3	6.5	10.7	17.5	20.8	22.6	21.9	19.8	14.3	7.5	1.7	(33)
		MCDB	1.2	1.7	9.9	16.0	21.4	24.6	27.2	25.8	23.3	17.4	9.2	2.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.6	11.6	12.5	13.2	12.6	13.1	12.3	13.3	12.1	10.3	8.5	(35)
		MCDBR	10.9	12.0	16.9	18.3	17.4	15.9	15.1	14.3	16.0	16.6	15.5	9.6	(36)
	5% WB	MCWBR	9.3	9.1	10.6	10.3	8.3	7.2	7.0	6.6	8.5	10.2	10.8	8.3	(37)
		MCDWR	10.1	11.7	14.5	15.6	14.4	12.8	12.8	11.8	13.0	14.2	14.1	8.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.261	0.308	0.327	0.363	0.396	0.403	0.396	0.393	0.362	0.325	0.292	0.267	(40)
		taud	2.398	2.256	2.372	2.336	2.292	2.305	2.332	2.333	2.416	2.511	2.526	2.448	(41)
	Ebn at Noon	Ebn at Noon	879	877	911	900	876	868	869	859	863	857	838	837	(42)
		Edh at Noon	74	103	106	119	128	127	123	118	101	80	66	65	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.68	2.76	3.88	4.53	5.41	5.96	6.08	5.22	4.18	2.66	1.71	1.25	(44)
	RadStd	RadStd	0.20	0.26	0.43	0.44	0.41	0.43	0.30	0.34	0.33	0.29	0.17	0.13	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page